Native-Asset Authorization Quantum Security and Resource Bound

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A native quantum chosen-request reduction, a concrete AES-128 resource assessment and implementations for EVM and Solana vaults.

1. Result and conditions

Fresh unauthorized release reduces to the fixed authority-preimage game, the derivation and private-generator games, and an explicit native proof-loss bound. The simulator answers adaptive classical authorization requests. The extractor uses the same quantum-hash experiment and one final database measurement.

> **Preliminary NIST Category 1 resource assessment**
>
> At the AES-128 reference budgets, the complete ideal proof-loss envelope is at most 0.325523274. The corollary uses the batch response-price condition c ≥ 1,048,576 gate-depth and a combined authority-preimage, derivation and private-generator advantage below 0.174476726. Published circuits and native fixed-prefix implementations calibrate the selected reference price in Section 3.

Quantity	Result
Code domain and FRI checks	4,096 positions, 64 distinct checks
Extension degree and committed degree	8 and 256
Joint native query density	$2^{-154.94110}$
Finite field-error density	$2^{-176.20344}$
Native proof loss at T = 2^{24}	≤ 6.42279849e-32
Native proof loss at T = 2^{64}	≤ 7.76107964e-8
Quantum AES-128 reference	G·D = 2^{170}, ideal success ≤ 3.25523274e-1
Classical reference at unit hash-call price	2^{143} gates, ideal success ≤ 2.54315058e-4

$$\Pr[\mathsf{ReleaseForge}] \le \mathrm{Adv}_{\mathrm{pre}}(\mathcal{B}) + \varepsilon_{\mathrm{der}} + \varepsilon_{\mathrm{rng}} + L(G, D, S, T).$$

The preimage adversary includes the complete deterministic source-decoding and candidate-validation work. The host supplies authenticated execution, protected persistent state and the canonical nonce policy. The exact primitive and host games appear in Appendix A.

2. Construction and proof geometry

A Parano1d secret derives an authority branch for each network and account index. A 24-element Poseidon2 permutation with eight full and 26 partial rounds maps fourteen private field elements and ten fixed capacity elements to twelve public outputs. Their 32-byte commitment identifies the vault authority.

The request is bound by SHA-256 and Keccak-256 on the same canonical encoding. One component enters the AIR. The second enters the transcript's initial state. Every Merkle root compares the complete 64-byte hybrid digest. The EVM job and cache, and the Solana base and request, retain both binding components.

The 128-row logical trace executes on its odd rows. Its 64 free logical rows and 128 hiding-PCS rows supply 192 private state-mask coordinates. Main transport uses at most 144. Quotient transport uses 72 of 128. The source domain remains 4,096 positions.

The query vector is a 64-element subset. One 512-bit response contains the twelve-bit grinding predicate and a 500-bit rank source. Cofactor reduction and colexicographic unranking select the subset. The exact finite source threshold is 896 good positions.

$$M = \binom{4096}{64}, \qquad \kappa_q = \frac{\lceil 2^{500}/M \rceil \binom{896}{64}}{2^{512}}.$$

$$\kappa_f = \frac{\lceil 2^{512}/p^{8} \rceil N_f}{2^{512}}, \qquad \kappa = \kappa_q + \kappa_f.$$

Each extension challenge likewise uses one full response, mapped to eight radix-p digits. The native compiler can therefore apply its one-local transition bound to the complete response. Five finite Reed–Solomon certificates and the exact fold identities supply the source ledger.

3. AES-128 resource comparison

NIST Category 1 uses AES-128 key search as its reference. The quantum calculation evaluates G·D = 2^{170} at depths 2^{40}, 2^{64} and 2^{96}. The classical calculation uses 2^{143} gates. At most 2^{64} classical authorization requests are included. NIST security evaluation criteria.

For each coherent batch, A is its logical gate count, δ its depth and k its number of charged responses. The minimum-cost condition is k·c ≤ A·δ with c ≥ 1,048,576 gate-depth. It applies to the cheaper hash component, including parallel execution and workspace reuse. The conservative adversarial physical-call count is bounded by G.

The resource assessment evaluates published direct-call layouts and native fixed-prefix circuits. Its native counting convention charges each live XOR and AND once and uses AND depth as an optimistic depth charge for node-by-node realizations. The all-batch contract states the condition for applying the theorem to other implementations. The exact evaluator gives a sufficient response price of 682,672 gate-depth for the ideal envelope below one half.

$$L = \varepsilon_{\mathrm{sim}} + \left(\sqrt{\frac{10 G D \kappa}{c(1-\beta/N)}} + E_{\mathrm{protect}} + E_{\mathrm{updates}} + E_{\mathrm{record}} \right)^2.$$

Depth D	Gates G	Complete ideal upper bound
2^{40}	2^{130}	3.25523274e-1
2^{64}	2^{106}	3.25523274e-1
2^{96}	2^{74}	3.25523274e-1

$$GD_{1/2}^{\mathrm{main}} = \frac{c}{20\kappa} \simeq 2^{170.619167401}.$$

The reference-budget rows include simulation, global prefix protection, retained updates and terminal recording. The dominant-term floor describes the main term. The full corollary uses the complete reference-budget envelope and the primitive-advantage margin in Section 1.

Batch response price c	Ideal upper bound at G·D = 2^{170}
1	1.00000000e+0
524,288	6.51046548e-1
1,048,576	3.25523274e-1
16,777,216	2.03452047e-2

The counted-query theorem gives ideal proof loss at most 7.76107964e-8 at T = 2^{64} physical hash calls. It uses the query budget directly. The gate-depth corollary additionally uses the batch response-price condition.

Native circuit resource assessment

The shortest candidate is a 64-byte transcript seal followed by a canonical four-byte field nonce. The first SHA-256 compression state is cached. The remaining compression includes its message schedule and digest feed-forward. Keccak-256 uses one padded 24-round permutation. The circuit checker specializes three fixed classical prefixes and 31 variable nonce bits, then removes unused output cones.

Lowest sampled layout	Counted data operations	Optimistic G·D
Keccak-256, full digest	35,440 AND + 105,762 XOR AND depth 24	3,388,848
Keccak-256, twelve grind bits	33,981 AND + 101,901 XOR AND depth 24	3,261,168
Keccak-256, one-bit XOR output	32,799 AND + 98,568 XOR AND depth 24	3,152,808
Keccak-256, one-bit phase data	32,703 AND + 98,360 XOR AND depth 23	3,014,449
SHA-256, one-bit ripple-adder output	18,321 AND + 74,164 XOR AND depth 1497	138,450,045

The one-bit phase implementation expands the output XOR cone into a parity frontier. Its 3,014,449 data product omits the CZ and Z phases, workspace cleanup and fanout. This optimistic same-layout charge is 2.875 times the selected reference price. Each row describes the sampled construction and its stated counting model.

The symbolic Keccak circuit reproduces 38,400 AND operations at AND depth 24. The four-T temporary-AND mapping agrees with the published clean-XOR construction, including measured cleanup. Native digests and projected outputs agree with SHA-256 and Keccak-256 reference implementations. Häner and Soeken, Meuli et al..

Classically prepared tables are charged during construction. The assessment separately records coherent prefix preparation, partial outputs, shared computation and quantum-memory strategies. Its resource model, published circuit ledger and native circuit counts specify the evaluated layouts and the cryptanalytic conditions for applying the Category 1 comparison.

4. Implementations and public withdrawals

Local execution

Both vault implementations complete native-asset deposits and withdrawals with the selected profile. The lifecycle checks cover altered inputs, changed recipients and amounts, stale requests, replay, independent relayers and recovery after a lost response.

Measurement	EVM	Solana v1
Authorization transactions	7	91
Execution charge	71,160,950 gas total	1,353,196 CU maximum
Maximum EVM transaction	11,507,479 gas	1,400,000 CU limit
Ordered verification transactions	7 including initialization	12
Temporary proof-account reserve	Contract state	0.343372 SOL, fully refunded

The current wallet run measured 18.75 ms median proving across 9 fresh proofs after 2 warmups, on an Intel Core i7-1365U laptop. Background load and temperature were not controlled.

Comparison with the published research

The 8 November article supplies the published local-withdrawal baseline. The current column contains the selected local profile measured on 9 November.

Measurement	Published	Current
Wallet proof generation	53.03 / 54.39 ms	18.75 ms median
EVM authorization transactions	10	7
Total EVM gas	103,890,467	71,160,950
Solana v1 authorization transactions	87	91

Measurement	Published	Current
Solana ordered verification transactions	14	12
Maximum Solana verification CU	1,140,062	1,353,196

EVM execution uses 31.5% less gas and three fewer transactions. Solana uses two fewer ordered verification transactions. Its complete fixture uses four more transactions and a higher peak CU charge.

The published wallet values are individual CPU samples. The current value is a nine-proof median on an Intel Core i7-1365U laptop. Both measure proof generation with public preprocessing and trace construction timed separately.

Signed Solana v1 wires are checked separately from the emulator's instruction replay. Solana accumulates each subset-rank term in its existing verification windows. Final rank equality is enforced before payout. This preserves the native acceptance conjunction while keeping preparation within the CU limit.

Public testnet execution

The selected profile completed withdrawals on Base Sepolia and Solana Devnet. An independent RPC audit checked signed transactions, deployed code, all 64 query positions, request binding and recipient balances. Each live runner also checked nonce consumption. Both Solana cases returned their complete temporary-account reserve.

Observed run	Transactions	Execution charge	Confirmed withdrawal
Base Sepolia	7	71,251,369 gas total	16.81 s
Solana Devnet v1	91	1,346,251 CU maximum	23.01 s
Solana Devnet v1, packet loss and reordering	92	1,364,077 CU maximum	25.21 s

These are individual network observations from a QEMU Virtual CPU version 2.5+ relayer host. The timer starts before proof preparation and ends at observed withdrawal confirmation. Deployment and deposits precede it. The public proofs took 31.97 to 41.60 ms on that host. The local laptop median above is a separate measurement.

Solana delivery uses direct validator TPU QUIC with four concurrent upload workers and twelve ordered verification windows. The faulted run recovered a deliberately missing chunk after packet reordering. Its network fee was 0.000465 test SOL, versus 0.000460 test SOL in the normal run. The normal temporary reserve was 0.25289764 test SOL and was fully returned.

The Base total authorization fee was 0.000427672327 test ETH. Its receipts record execution and L1 data charges separately, with zero operator fee. Confirmation and finality observations retain their network-specific labels in the evidence. The Solana audit checked finalized data for every captured transaction.

Public withdrawals: Base Sepolia, Solana Devnet normal and Solana Devnet with packet loss. The public-run summary binds these values to the raw receipts and independent audits by SHA-256.

5. Reproduction

The reading report embeds all proof texts below. Exact arithmetic and source fingerprints are in security-evaluation.json. The finite witnesses are in rs-certificate.json. Code and execution fixtures are under experiments/native-vault. The circuit checker uses PyCryptodome. The reproduction guide lists dependencies and additional transcript checks.

```
python3 research/security/check_rs_certificate.py
python3 research/security/check_profile.py
python3 research/security/check_mask_transport.py
python3 research/security/check_sampler.py
python3 research/security/check_hash_circuits.py
python3 research/security/evaluate_security.py
python3 tools/summarize_optimization.py
python3 tools/summarize_public_runs.py
```

Proofs

The following appendices give the game, source theorem, native simulation and resource conditions for this exact implementation.

Appendix A. Authorization game and corollary

A canonical request is authorized by knowledge of a reusable authority secret. Its proof binds the request to the deployed vault, parameters, recipient, amount, nonce and expiry. The proof checker uses KoalaBear, a degree-eight extension, a 4096-position domain and 64 FRI checks.

1. Authority relation and public instance

The secret has fourteen elements of F_p. Poseidon2 has width 24, ten fixed capacity elements, a cubic S-box, eight full rounds and 26 partial rounds. Its first twelve output elements are Y. The installed authority is C=SHA256(AUTH_DOMAIN||encode(Y)). Field and integer encodings are canonical. The exact fixed capacity and constants are source-linked.

For the security game, Y is supplied as public auxiliary input at setup. It is the same information carried by an honest authorization capsule. The adversary receives the installed C, parameters and this Y. It may request classical authorizations adaptively and query the two public hash components coherently.

The authority-preimage game gives this same public instance. It accepts any secret whose computed output Y' has commitment C. This definition covers both the fixed Poseidon2 preimage and a substitute output mapping to the installed commitment. The secret distribution is the protocol's actual HKDF and reduction distribution.

2. Fresh-request game and host policy

Let I=SHA256(M)||Keccak256(M) for the canonical request M. The requested set J records I before any response is constructed. A forgery is a value release for I outside J. Observed responses may be chosen adaptively. The host checks the installed verifier, authority, parameter manifest, recipient, available value, expiry and current nonce. It protects all persistent verification state and consumes the nonce before transfer. The network execution and host history are authenticated.

Repeated submission of an already authorized request is governed by its nonce. The cryptographic game and the host policy apply to the same canonical request. The application may expose this authorization interface outside a blockchain whenever it supplies the same policy checks and authenticated execution.

3. Native reduction

The joint HKDF and private-generator hybrids contribute eps_der and eps_rng. The witness-free simulator then supplies adaptive responses with event error eps_sim(S,T), using the actual native transcript and one retained permutation of private leaf inputs. Its field masks and source words are simulated from public Y and I.

The source extractor and the protected-response compiler establish

```
Pr[ReleaseForge] <= Adv_pre(B)+eps_der+eps_rng+L(G,D,S,T).
```

Here L is the exact expression in compiler.txt. The same simulated experiment supplies both the simulator and the final extractor. Every component, constructor, setup and final-pair call is counted in T.

B has the original adversary's quantum execution followed by one compressed-database measurement. It enumerates at most T rooted prefixes. Each source reconstruction uses the deterministic first-order DKT26 decoder at the finite parameters in rs-certificate.json, checks all returned source candidates on the 128-point AIR domain and validates any recovered authority secret against C. The full classical decoding and candidate-validation work is included in Adv_pre(B). The primitive condition is stated for this complete B, including that work.

4. Category 1 corollary

For at most 2^{64} classical authorizations, reference gate-depth GD=2^{170} and depths D in {2^{40},2^{64},2^{96}}, impose the exact batch resource condition of resource-contract.txt: c_j>=2^{20}. Every adversarial call is conservatively charged one logical gate. The reproducible complete ideal proof-loss envelope is at most 0.325523273624783. At 2^{143} classical gates it is at most 0.000254315057964 even charging one unit per hash call.

Consequently the full authorization game has success below one half at both Category 1 references whenever

```
Adv_pre(B)+eps_der+eps_rng < 0.174476726375217.
```

This gives the NIST Category 1 corollary under the displayed primitive, resource and host conditions. Its dominant ideal gate-depth half-success floor is approximately 2^{170}.619167401. The complete finite terms are included in the reference-budget bound, rather than omitted from that floor's interpretation.

The finite proof, its native response geometry and both host verifiers are linked by the reproduction checks in this directory. The completed assessment in nist-resource-assessment.txt supplies concrete resource evidence from published and native fixed-prefix circuits. It states the evaluated call family and the broader batch-cost and primitive conditions for applying this corollary to a chosen adversary circuit model.

Appendix B. Algebraic simulation

The polynomial opening view admits an efficient witness-free simulator outside an explicit set of exceptional verifier coins. Its proof uses affine translations of the actual main, quotient and randomization masks.

1. Model and notation

Let F = F_p, p = 2130706433, and E/F have degree eight. Private field draws are independent and uniform in this section. Verifier challenges are fresh uniform extension-field elements. Commitment digests and authentication paths belong to the native hash compiler in compiler.txt.

The logical AIR has 128 rows and transition stride two. Its execution rows are 1,3,...,127. In the committed order-256 domain these are the coset where X^64=-1. Let V(X)=X^64+1. The remaining 192 values per private state column are independent uniform masks: 64 explicit free logical rows and 128 hiding-PCS rows. Therefore

```
T_j(X)=U_j(X)+V(X)r_j(X), deg U_j<64, deg r_j<192.
```

This follows from the evaluation isomorphism on 256 distinct points. The request columns encode public context and can be coupled identically. The execution rotation g has order 64. Quotient chunks have 128 fresh mask coordinates and committed degree below 256. Randomization columns have degree below 256. The common code domain still has 4096 points. The AIR divisor is Z_H=X^128-1. Its actual padded active selector is A=(1-X^64+X^128-X^192)/4. Hence A*V=-(X^256-1)/4, a multiple of Z_H. Every execution selector vanishes on the free logical rows. Capacity is constrained at row one. Last-row output constraints apply at row 127.

Use alpha for the AIR-composition challenge, z for the out-of-domain point, gamma for the PCS batching challenge and beta_i for FRI folding challenges. Let Q be the set of distinct queried source points. There are 64 query positions, so |Q| <= 64. The proof opens main columns at z and gz, quotient columns at z and randomization columns at z.

The opening view consists of all disclosed base and extension values, the final FRI polynomial and every disclosed FRI row. In particular a FRI row of arity four contains four aggregate values. Each input batch opens one source row at each queried source location. Uniform simulation of the complete aggregate polynomial handles all four FRI row values.

2. Interpolation and Frobenius orbits

For a in E, let O(a) be its Frobenius orbit over F. Its cardinality is the degree of its minimal polynomial m_a. Values of an F-polynomial at O(a) must be Frobenius conjugates. Conversely, arbitrary compatible values at distinct orbits and base-field points interpolate to an F-polynomial whose degree is below the sum of the orbit degrees. This follows from the Chinese remainder theorem for their pairwise coprime minimal polynomials, or equivalently from interpolation over the splitting field.

If a != 0, then O(a) and O(ga) are disjoint. Otherwise a^(p^j)=ga for some 1 <= j < 8. Iterating the j-th Frobenius 8/gcd(j,8) times gives g^(8/gcd(j,8))=1. This contradicts the order 64 of g. For a=0 the two points coincide and impose the same evaluation condition.

3. Main-column transport

Take two valid witnesses for the same public (Y,I). Fix alpha, z, gamma, the FRI challenges and Q. Exclude z in the public trace or code domains where the following evaluation denominators can vanish. These public domains and their g-rotations contain at most 8192 points.

Couple the two masked main traces to agree on

```
Q union gQ union O(z) union O(gz).
```

This requires at most 2*64+2*8=144 base-field coordinates. At these points V is nonzero. Interpolate the required change in r_j after division by V. It has degree below 144, hence below 192. Adding this fixed interpolant is a bijection of the uniform main-mask space.

All main observations are equal under this coupling. Both local and rotated trace values needed to evaluate constraints agree at Q and O(z). The implementation's conservative runtime budget is

```
    64 + 8*2 = 80 <= 128 for direct PCS disclosures.
```

The state transport uses 144 <= 192 coordinates. Each quotient mask uses 72 <= 128. These separate inequalities justify both the runtime PCS guard and the additional AIR transport.

The request columns are public and identical between the two witnesses.

4. Quotient transport

Each individual valid AIR-constraint polynomial C_l(X) is divisible by Z_H. Expand the combined quotient in a fixed F-basis e_b of E:

```
    sum_l alpha^l C_l(X)/Z_H(X) = sum_b e_b Q_b(X).
```

Its basis coefficients are F-polynomials. The transported main values imply that each Q_b agrees between the two witnesses at Q union O(z). This conclusion follows by expanding every alpha^l in the fixed basis before evaluation. It treats all disclosed quotient-basis openings.

There are eight disjoint quotient cosets H_i of size 128. Write V_i for their monic vanishing polynomials and

```
    L_i = c_i product_(j != i) V_j.
```

The unmasked Lagrange chunks q_i satisfy sum_i L_i q_i = Q_b. Their masks are

```
    q'_i = q_i + V_i t_i,                 0 <= i < 7,
    q'_7 = q_7 - V_7 sum_(i<7) (c_i/c_7)t_i,
```

with independent uniform F-polynomials t_i of degree below 128. The added terms cancel in the weighted sum.

At each observation point outside all H_i, choose changes in the first seven t_i to cancel the first seven chunk differences. The equality of the combined Q_b forces cancellation in the last chunk. At a point of H_i the corresponding chunk already agrees because its selector is one. The other seven chunk values are adjustable. If i<7, its unused t_i value supplies the remaining last-chunk adjustment. If i=7, the last mask vanishes and the other seven adjustments are free.

These adjustments are Frobenius compatible. Interpolating them requires at most |Q|+8 <= 72 coordinates per independent mask. Their degrees remain below 128. Conditional on the main masks, each adjustment is a translation of the uniform quotient-mask space. It preserves the full joint distribution, including the balancing relation for chunk seven. Extra random padding columns can be coupled identically.

5. Complete FRI aggregate transport

Let W be the initial DEEP batched polynomial. A source column P opened at a contributes a known power of gamma times

```
(P(X)-P(a))/(X-a).
```

All source columns have degree below 256, so deg W < 255. Couple the randomizer's disclosed observations identically at first. Let Delta W be the remaining aggregate difference from the transported main and quotient polynomials.

The coupled direct observations make Delta W vanish at Q. It also vanishes at every point in O(z) except z. To see this, evaluate each source difference separately. Its numerator vanishes at the entire relevant orbit. For a=z it still has the factor m_z(X)/(X-z). For the main opening a=gz, orbit disjointness supplies this same factor. The case z=0 has no additional orbit point to check.

Consequently

```
Delta W(X) = A(X) B(X),
A(X) = Z_Q(X) m_z(X)/(X-z),
deg B < 256-|Q|-deg(m_z),
```

where Z_Q is the monic product of X-q over distinct q in Q.

The randomization batch contains sixteen independent F-polynomials of degree below 256. The first eight are batched with powers 1,gamma,...,gamma^7. If gamma has degree eight over F, these powers form an F-basis of E. Decompose

```
-B(X) = sum_(j=0)^7 gamma^j B_j(X),   B_j in F[X].
```

Translate these eight randomization columns by

```
R_j(X) <- R_j(X) + Z_Q(X)m_z(X) B_j(X).
```

The degree remains below 256. Every randomization observation at Q and z is preserved. The aggregate change is -Delta W, because the DEEP denominator is X-z. Hence the complete initial aggregate W, all FRI folds, every opened FRI row and the final polynomial agree exactly.

The translations are triangular: the main translation is fixed, quotient translations may depend on the main masks, and the randomizer translation may depend on both. Each step is conditionally bijective on its uniform mask space. Their composition preserves the full product distribution.

6. Efficient simulation

Sample the honest verifier coins and the query positions. Sample the main values at Q union gQ union O(z) union O(gz) from their uniform Frobenius-compatible evaluation spaces. These spaces are independent of the witness because the main-mask evaluation map is onto.

Evaluate each constraint quotient at Q union O(z) from those main values and the public preprocessing. Sample quotient-chunk observations uniformly in the affine fibers of the corresponding Lagrange equations. The translations in Section 4 show that these are their exact conditional distributions. Sample the randomizer observations and public-padding observations with their prescribed distributions.

The disclosed source values determine W at Q and O(z)\{z}. Conditional on those values, the randomizer transport spans the complete kernel of these evaluations. Thus W is uniform in the affine space of E-polynomials of degree below 255 with those values. Its remaining dimension is

```
255-(|Q|+deg(m_z)-1) = 256-|Q|-deg(m_z) >= 184.
```

Interpolate one such W, add a uniformly random kernel polynomial, and compute its complete FRI fold sequence. Return its queried rows and final polynomial together with the sampled direct observations. Every step is finite-field linear algebra of polynomial size. All inverse hints are determined by public challenges and query points.

This simulator establishes the distribution of the full algebraic opening view from public (Y,I), including all quotient coordinates and all FRI leaf values.

7. Exceptional coins and repeated use

Every proper subfield of E lies in $F_{(p^4)}$. A uniform gamma therefore fails the degree-eight basis condition with probability exactly p^{-4}. The excluded z-domain probability is at most $8192/p^8$. Taking a union bound gives

```
delta_alg <= p^-4 + 8192/p^8,
-log2(delta_alg) approximately 123.95474.
```

Conditional on the complementary event, simulation is exact. The unconditional total variation distance is at most delta_alg. S fresh independent-coin authorizations have distance at most S*delta_alg by successive conditional simulation. Classical adaptive request selection is permitted in this algebraic honest-verifier experiment.

The implementation supplies seeded coins and hash-derived verifier challenges. The joint generator hybrid and native adaptive quantum compiler supply the additional terms for that experiment. Twenty uniform base-field salt elements have entropy 20*log2(p), approximately 619.77 bits, in the independent-coin hybrid.

8. Finite checks and references

check_profile.py verifies a Proth primality certificate for p, Rabin's irreducibility criterion for X^8-3, both linear-layer ranks, bijectivity of the cube map on F, the order of g, proper-subfield dimension, source opening counts and the finite interpolation budgets. finite-profile.json records exact rational values and source fingerprints.

The Lagrange mask construction is described in Ulrich Haboeck and Al Kindi, A note on adding zero-knowledge to STARKs, Section 4.2: https://eprint.iacr.org/2024/1037

The argument above includes the disclosed quotient basis coefficients, the rotated main opening and complete aggregate-polynomial transport for this profile.

9. Causal algebraic sampling

The algebraic simulation can also sample its messages in verifier order. At z, sample the main and quotient observations in the spaces described above. Sample each R_j(z) uniformly in F(z), with the appropriate Frobenius compatibility. Repeated opening points impose one shared value. After gamma, consider the aggregate with no

source query fixed yet. The randomizer transport with |Q|=0 shows that W is uniform among degree-below-255 E-polynomials with its determined values on O(z)\{z}. This conditional law depends only on the disclosed observations.

Sample that complete W. Each subsequent beta_i determines its next fold and eventually the final constant. After the source queries are sampled, draw their main and quotient observations conditioned on the earlier z-observations. The complete W is independent of their undisclosed polynomials given those earlier observations, by the same surjection of the randomizer masks.

For a queried base point q, write W_other(q) for the DEEP contribution of the main, quotient and public preprocessing rows. The randomizer row must satisfy

```
sum_(j<16) gamma^j R_j(q)
  = sum_(j<16) gamma^j R_j(z)
    + (q-z) [W(q)-W_other(q)].
```

This is a surjective F-linear equation from F^16 to E. Choose the last eight coordinates uniformly in F and solve uniquely for the first eight using the gamma basis. It gives the uniform affine fiber at q.

These fibers are jointly compatible with the earlier R_j(z) and complete W. Indeed the kernel of the coefficient map

```
(u_0,...,u_15) -> sum_j gamma^j u_j
```

has F-dimension eight. Multiply polynomial vectors in this kernel by m_z(X). Their evaluations cover the entire kernel independently at Q, because |Q|+deg(m_z) <= 72 < 256. Thus this sampling law is the exact conditional distribution of every disclosed randomizer row.

The construction samples the algebraic messages causally. simulation.txt and compiler.txt establish the native commitments and continued quantum experiment, including the complete buffered response groups.

Appendix C. Finite source restoration

The fixed 4096-position profile has a polynomial-time prefix extractor with round-by-round error

```
kappa_public <= C(896,64)/C(4096,64) + N_f/p^8,
p=2130706433.
```

The query term has negative binary logarithm 142.94109606984. The exact finite first-order certificates and integer N_f are reproduced by check_rs_certificate.py. This analysis uses 64 queries, the existing blowup sixteen and the existing degree-eight challenge field.

1. Immutable words and finite interpolation certificates

Expose committed batches as immutable arrays. Keep every prover message and verifier coin in its native order. Main words precede alpha. Quotient and randomization words precede z. Opening claims precede gamma. Each current FRI word precedes its folding challenge. The final constant precedes the complete 64-position query move.

Use the finite first-order interpolation and counting results of Dao, Kominers and Thaler, 2026/2056, Proposition C.2 and Lemma 5.5. They apply to arbitrary prescribed evaluation domains. The characteristic conditions are checked below for the actual prime, including when source columns are encoded in a larger extension for interleaved list decoding.

For length n, dimension k, threshold A, multiplicity m, derivative degree M and total jet degree B, put D=k-1. The derivative-weighted support has monomials X^x Y_0^a Y_1^b satisfying b<=M, a+b<=B and x+D*a+(D-1)*b<m*A. Its degree-t coefficient count and local rank bound are

```
N_t = sum_(b=0)^min(t,M) max(m*A-D*t+b,0),
R_t = sum_(s=0)^(m-1) 1_[s+(D-1)t<m*A]
      * max(0,min(m-s,min(t,s)-max(0,t-M)+1)).
```

Put C=sum_t N_t and R=n*sum_t R_t. Every certificate has C>R and

```
(C-R)*H > R*B.
```

For a received curve of degree ell, this implies C*(ell*H+1)>R*(ell*H+ell*B+1). A coefficient count therefore gives a nonzero polynomial kernel vector of challenge degree at most ell*H. Dividing its common univariate content makes its specialization nonzero at every challenge. Substituting any degree-<k candidate with A agreements then gives a polynomial of degree below m*A and at least m*A roots, counting multiplicity. It vanishes identically. The support rank bounds are Proposition C.2. This argument also explains why the line-to-affine reduction is not used for powers batching.

The checked tuples are listed as (n,k,A,m,M,B,H).

```
Source list:    (4096,257,896,256,157,888,405516956)
Gamma and FRI:  (4096,256,896,192,126,672,177710)
FRI:            (1024,64,224,64,42,227,53572)
FRI:            (256,16,56,24,15,87,2838)
FRI:            (64,4,14,4,2,18,390)
```

The Gamma tuple is used again for the first fold. All five finite certificates satisfy p>max(D,M). The largest derivative degree is 157. Every folding threshold equals ceil((7/32)*n).

2. List and curve bounds

Put lambda=(n-D)/(A-D), b=B*(2M+1) and h=H*(2M+1). The squarefree first-order counting theorem gives the following conservative bounds:

```
Lambda = ceil(4*D*lambda*B*M+2*B*M+B),
E = ceil((2b-1)h+lambda*(h+b+4*D*b*h)+(n-D-1)b
    +(48*D^2*H+16*D)*lambda^2*B*M
    +8*D*(n-D-1)*lambda*B*M).
```

Lambda bounds every fixed-word list. A degree-ell received curve has at most ell*E exceptional challenges. To see the latter specialization, use challenge degree ell*H in Lemma 5.5. Its ordinary-resultant and regular-family terms are bounded as in the proof of Theorem 5.2. Each term is linear in the received-curve degree and this height, giving the displayed ell*E envelope. For M=0, Remark 5.6 supplies the ordinary case and the regular terms vanish.

At every other challenge, every candidate with A agreements admits coefficient witnesses on its full agreement set. The counting proof splits ordinary resultant solutions from regular differential solutions. Taylor reconstruction uses p>D, and separability uses p>M. These are exactly the characteristic guards checked for each tuple. Corollaries 6.4 and 6.6 transfer the list and curve statements to interleaved words without a width factor.

Let Lambda_src be Lambda for the source-list tuple. This bound is independent of the received word and field extension. Lists are decoded by the first-order algorithms in Section 9 of the same paper. For an interleaved source, encode all columns in an extension of the base field, decode, and expand the coefficients in its fixed basis. The algorithm is polynomial in the code length, field representation and certificate parameters.

3. Restoration through the actual FRI folds

Each production domain is a multiplicative coset and each arity r is four. Write t=x^r. Fourier decomposition of a current word v defines coefficient words u_j(t) by

```
v(x)=sum_(j=0)^(r-1) x^j u_j(x^r).
```

The actual native fold at beta is sum_j beta^j u_j(t). The field-matrix checker compares this identity with the optimized two- and four-point formulas in the streaming verifier. It checks every coordinate basis, so the same linear identities hold over the extension field.

Suppose the next restored polynomial Q(t), of degree below k/r, agrees with the folded word at the descendants of a good query set. Pull it back to Q(X^r). On the current domain its full agreement set S contains the good set and is a union of complete r-point orbits. Apply polynomial- curve MCA to the words u_j(X^r), at degree r-1, using the current tuple. Outside its exceptional set, it gives polynomials U_j(X) of degree below k agreeing with those words on the full set S.

For a primitive r-th root omega, U_j(omega*X)-U_j(X) vanishes on S. Its degree is below k, while |S|>=A>k. Therefore it is zero. Since r is invertible, U_j(X)=Q_j(X^r) with degree Q_j<k/r. The polynomial sum_j X^j Q_j(X^r) has degree below k and agrees with v on S. This restores the previous word while retaining every original good query. The full-set property supplies the orbit closure used in this step.

Starting with the transmitted final constant, repeat through arities 4,4,4,4. The code lengths and dimensions are (4096,256), (1024,64), (256,16) and (64,4). The initial FRI coordinate is unshifted. Pulling its polynomial back by the public source coset shift gives the same degree bound and agreement set on the source domain.

4. Direct restoration of opening claims

Before gamma, each source column P_j and its advertised claim v_j at z_j are fixed. Its DEEP word is

```
d_j(q)=(P_j(q)-v_j)/(q-z_j).
```

Exclude z values where a public evaluation denominator vanishes. The public domains and rotations cost at most $8192/p^8$. The native batching word is a powers combination of 268 such words, of curve degree 267. Apply its finite Gamma certificate to the restored initial FRI polynomial. It supplies degree-<256 polynomials R_j agreeing with every d_j on the same full set, outside at most 267*E_gamma challenges.

Set P'_j(X)=(X-z_j)R_j(X)+v_j. It has degree at most 256, agrees with the corresponding committed source on that set and takes the advertised value at z_j. Two reconstructions of a main column, at z and g*z, agree on more than 256 points. Hence they are identical and both claims are its evaluations. The public preprocessing reconstruction likewise

equals its authenticated polynomial.

Each private source value on this set is in F_p. The coefficientwise Frobenius difference of P'_j has degree at most 256 and vanishes at more than 256 base-field points. It is zero, so P'_j has coefficients in F_p. The reconstructed source tuples therefore belong to the lists of dimension 257 fixed before their respective verifier coins.

5. AIR flags and exact field ledger

For every decoded main candidate, form all constraint remainders modulo Z_H. An invalid trace has a nonzero remainder. Its alpha composition is a nonzero polynomial of degree at most 61, giving at most 62*Lambda_src exceptional alpha values.

Outside this flag, pair each main candidate with every decoded quotient candidate. The composed constraint polynomial minus the reconstructed Z_H multiple is nonzero. The native outer equation evaluates this polynomial at z. Its degree is at most 4608. This bound includes main and quotient source degree 256, the degree-896 chunk selectors and Z_H of degree 128. Thus all pairs cost at most 4608*Lambda_src^2 exceptional z values. Since Lambda_src^2>=8192, the excluded public-domain points fit the additional Lambda_src^2.

The complete field numerator is

```
N_f = 62*Lambda_src+4609*Lambda_src^2
      +267*E_gamma+sum_i (r_i-1)*E_i.
```

Every integer and characteristic guard is reproduced in rs-certificate.json. The four E_i use their own finite tuples. All constructor, field, commitment and verifier parameters are those of the fixed implementation profile.

6. Query bound and round-by-round knowledge

Fix a complete transcript before its final query move. Its good-index set contains exactly those source positions passing the complete native query predicate. If its density exceeded a=7/32 and all flags above were absent, backward folding and direct opening restoration would give source candidates on that same good set. The outer equation would then contradict the absence of the alpha and z flags unless a valid main trace were in the decoded list. The prefix extractor enumerates that list, checks the trace on the 128-point subgroup and returns its first fourteen elements at logical row one as an authority preimage.

Consequently, on failed extraction and with all flags absent, at most 896 of the 4096 source positions pass. A uniform 64-element subset is contained in that set with probability C(896,64)/C(4096,64).

The implemented query sampler uses one 512-bit response. Its low twelve bits must be zero. The remaining 500 bits, reduced modulo M=C(4096,64), select the subset by colexicographic rank. Every rank has at most ceil(2^{500}/M) preimages. Its exact joint bad-response density is at most

```
kappa_query = ceil(2^500/M)*C(896,64)/2^512.
```

Every field challenge is the radix-p expansion of a uniform 512-bit integer reduced modulo p^8. A set of b extension values has density at most ceil(2^{512}/p^8)*b/2^{512}. Thus

```
kappa_field = ceil(2^512/p^8)*N_f/2^512,
kappa = kappa_query+kappa_field.
```

The prefix extractor decodes the same finite source list, checks the 128-point trace with transition stride two and returns the secret at logical row one. It uses the deterministic first-order decoder of DKT26, Section 9. Its characteristic guard p>2*(B+1)^3 holds for B<=888. The scalar source support and kernel-height guards are checked exactly by check_rs_certificate.py. This is a polynomial-time extractor with fixed code length 4096. Its full deterministic decoding cost is included in the authority-preimage adversary in authorization.txt.

Appendix D. Native adaptive simulation

Let H=SHA256||Keccak256 be a uniform 512-bit random function, with coherent access to either component. Its leaf, node and transcript namespaces are the actual disjoint namespaces in compiler.txt. Put

```
N=2^512, n=13648, u=13641, v=n+u=27289, eta=2^-140.
```

Fix the public seed row H(AUTH_DOMAIN)=d. It determines the ten authority capacity elements. All other hash rows form a uniform completion. The seed row has length 40, disjoint from every leaf, node and transcript input used below. The simulator and every input exchange fix this row.

An attempt contains three private 4096-position source trees and four FRI trees of sizes 1024,256,64,16. It reveals at most K=448 private leaves. The public preprocessing tree is separately authenticated. Every private leaf contains at least sixteen payload elements and twenty independent field salts per matrix row. The quotient leaf concatenates eight separately salted rows. Its dummy-input domain has cardinality at least p^{36}. Actual-view leaf inputs retain twenty fresh salts. All encodings are injective.

1. A balanced random-function table

For each complete private leaf width b in {36,52,68,288}, let A_b be the set of its correctly encoded b field elements. For a fixed digest y, the fiber size |{r in A_k:H(r)=y}| is binomial with expectation p^b/N. The multiplicative Chernoff bound and a union over all y give failure at most

```
8 N exp(-eta^2 p^36/(3N)).
```

The node map has domain {0,1}^1024 and the same codomain. Its analogous balance failure is at most 2 N exp(-eta^2 N/3). Their sum is below e_load=2^{-1024}. Indeed $p^{36}>2^{1115}$, so the first exponential exponent exceeds 8192/3>2048, and the entire prefactor is below 2^{516}.

On this event every output fiber has size between (1-eta)|A|/N and (1+eta)|A|/N. A uniform dummy leaf input conditioned on its digest and the complete physical oracle is uniform in that fiber. Its point mass is at most $2N/p^{36}$.

Conditional on H, disjoint dummy subtrees use independent leaf coins. Induction from the balanced leaf and node maps bounds every subtree root's point probabilities by (1+eta)^w/N and (1-eta)^w/N, where w is its vertex count. For all private trees, (1+eta)^v<2. This provides the fresh-root entropy used below.

Conditioning additionally on the retained paths and their frontier labels preserves the leaf-input fiber bound for an unopened leaf. Its private dummy row affects the retained computation only through its leaf digest. All other dummy rows can be fixed when taking this conditional law. A full algebraic opening view and all honest field polynomials may also be fixed, because they depend on this dummy row only through the already fixed roots and retained labels.

2. Exact full-forest refresh

For an honest field tape, build a dummy physical tree with independent uniform field payloads and salts, and retain its actual H evaluations. Let r_i be its leaf inputs and y_i=H(r_i). Give the corresponding honest field row a fresh salt and call its encoded input x_i. Impose distinct, disjoint endpoints and preservation of earlier public pairs. Define

```
sigma_full = product_i (x_i r_i).
```

The logical oracle H composed with sigma_full hashes each honest row to y_i. Node inputs and transcript inputs are unchanged, so the dummy authentication tree is also the honest logical authentication tree.

The exact refresh identity is input-label invariance of a uniform random function. For two as-yet-unread input rows, exchange their full outputs and their input names. Both rows remain independent uniform rows. Apply this identity in the prover's causal order. A main word is fixed before its root, quotient and randomizer words before their roots, and each aggregate fold before its next root. The private dummy rows can therefore be read under physical H while honest rows use the exchanged logical names. Typed-domain separation fixes every node and transcript call. Endpoint and public-pair failures are charged below.

For a compressed database with t entries, exchanging fresh endpoints changes its state only on bases containing either endpoint. A fresh actual salt bounds one membership probability by t/p^{20}. A fresh dummy row bounds the other by t/p^{36}. The norm difference is at most twice the square root of their sum. Summing these cold exchanges gives the prefix comparison for arbitrary quantum prefixes and classical adaptive requests. Fresh salts are drawn after the request is fixed. The comparison conditions on each request register and uses raw, unnormalized tapes, so it introduces no guard-conditioning denominator.

3. Causal field views and native coins

At request start mark I in J. Commit the dummy main root and derive alpha. Commit dummy quotient and randomizer roots and derive z. Sample the out-of-domain observations, derive gamma and sample the complete aggregate W. Commit each next dummy FRI root, derive its beta and fold W. Absorb the final polynomial and derive all native query positions. Sample the direct source rows conditioned on those observations and W. Install only their retained leaf exchanges and emit the native capsule.

Use the causal sampler of masking.txt. Native alpha and z are derived after fresh independent dummy roots. The batching input contains the two main out-of-domain observations, whose mask evaluation space has at least eighty independent base-field coordinates. Each fold follows a fresh independent dummy FRI root. These inputs have conditional point mass at most 2/N, or p^{-80} at the batching move, before the corresponding private native query.

An absent-row compressed query differs from the ideal extension isometry by norm sqrt(2/N). Each next raw full-state output is uniform. The complete retained digest makes the next continuation input an injective function of that output. Against a t-entry database, a continuation's first old-input or self-return probability is at most (t+ell+1)/N. Keep all private outputs. Deleting the newly added rows recovers the initial database uniquely, so the estimate holds on an arbitrary coherent database state.

Every field challenge consumes one full response. Its radix-p map differs from uniform on $F_{(p^8)}$ by at most p^8/N in total variation. There are seven field challenges. The query sampler is simulated with its exact native subset distribution, so no uniform-subset approximation is used.

For all N possible sealing digests, the probability that some digest has no successful witness among its first 2^{24} candidates is at most $N*(1-2^{-12})^{(2^{24})}<2^{-5397}$. The candidate inputs are distinct reduced field encodings. Hence this single random-table event bounds every adaptive honest grind simultaneously. On its complement each constructor

uses fewer than $L=2^{26}$ physical calls including both component calls, its forest and transcript.

Fresh-response comparison and a union over those raw tapes give

```
b_coin(T)=(2^26+4096) sqrt(2^12*(T+v+256)/N)
            +7 p^8/N.
```

The output of the first passing witness has its grinding bits fixed and its other 500 bits uniformly distributed. Applying the same rank map preserves the native query distribution exactly. Failed trials remain charged calls. The global witness-prefix exception contributes 2^{-5397}.

This couples the honest field messages to the independent-coin algebraic sampler. Its independent-coin algebraic exception per attempt is

```
delta_alg=p^-4+8192/p^8.
```

4. Accessible-view pruning

Retain only the leaf exchanges for positions in the emitted capsule. Duplicate positions reuse one exchange. Omitted actual endpoints have point mass at most p^{-20} after fixing physical H, the complete field view, all honest field polynomials and the retained table. Omitted dummy endpoints have point mass at most $2N/p^{36}$ by Section 1. Thus the changed support after deleting omitted exchanges has point bound

```
delta=n*(p^-20+2N/p^36).
```

For a q-query reference circuit that uses the retained table, condition on physical H and its complete accessible view. Its query states contain no additional omitted-row data. The expected squared mass on the changed support is at most delta at every query. The ordinary oracle hybrid inequality consequently bounds its average state distance by 2q sqrt(delta). This argument uses accessible query registers, rather than the constructor's database-membership event. It therefore applies even though the constructor has already read the private dummy rows.

After pruning, the complete private field polynomials and unopened dummy rows can be erased. The retained rows have exactly the causal sampler's law. The functional simulator needs only public (Y,I), independent dummy forests, the causal field sampler and at most 448 input exchanges.

5. Finite guards and adaptive theorem

T charges every adversary, setup, constructor, component-projection and verifier-recheck call. In particular T>=v for a completed attempt and Sv<=T for S completed full-forest attempts. The following monotone envelope is evaluated at S<=floor(T/v)+1, allowing one final unfinished constructor on whole-experiment abort. For T>=v this also gives 896*S<T. Every numerical term is monotone in S.

Let

```
Phi(T)=64(T+v)^2/p^20 + 64(T+v)^2/N.
```

This union covers endpoint aliases, earlier dictionary keys, preservation of public leaf-computation pairs and fresh root labels hitting earlier public digest targets. All counts are taken on raw tapes. There are fewer than 2(T+v) earlier or proposed endpoints and fewer than v private vertices. Fresh salts give the p^{-20} term. The balanced-label bound 2/N gives the digest term. Public preprocessing calls are included in T.

The native simulator has event error bounded by

```
    eps_sim(S,T) <= min(1,
        S*delta_alg + 8ST sqrt(delta) + 4S Phi(T)
        + S*b_coin(T) + 4S*e_load+2^-5397).
```

Proof. Replace adaptive attempts in reverse order. Each replacement has a real prefix and an arbitrary continued experiment. Apply the exact refresh and cold-endpoint comparison, the native coin coupling and the algebraic sampler. Delete omitted exchanges with the accessible-view hybrid. For T>=v, the 8T sqrt(delta) term covers both the summed cold refresh and the 2T sqrt(delta) pruning cost with slack. Abort the whole simulation on a retained guard failure. Raw full-tape failure estimates cover the retained guard. Sum all errors, including the global balanced table event, over S. Later authorization responses use the composed retained input permutation, so the continuation is included in every replacement. The simulator is expected polynomial time and preserves the exact native transcript and all Merkle path checks.

The reduction replaces private seeded generators by independent uniform coins before applying this theorem. Their joint pseudorandomness loss is the separate eps_rng term in authorization.txt.

Appendix E. Protected atomic-response compiler

The native proof uses one 512-bit response for every extension challenge and one response for the complete 64-position subset and its grinding predicate. Its resource bound applies to classical authorization requests and coherent access to both hash components.

1. Native response grammar

Write H=SHA256||Keccak256 and $N=2^{512}$. Each observed message is appended to the complete previous digest. Field sampling consumes all 64 bytes. It reads their reverse byte order as a little-endian integer, reduces it modulo p^8 and returns its eight radix-p digits. Its largest point mass is ceil(N/p^8)/N. A field bad set of cardinality b consequently has probability at most b*ceil(N/p^8)/N.

The query seal hashes the final polynomial and its full predecessor. The candidate input is the seal digest followed by one reduced field word. Its output is one integer y in [0,N). Accepting a witness requires y mod 4096=0. The subset rank is floor(y/4096) mod C(4096,64). Thus the source density is exactly the envelope in source-restoration.txt. All trials are ordinary charged calls. A transcript with an exceptional out-of-domain challenge is charged at the corresponding field flag.

Merkle inputs have fourteen-byte leaf and node tags. Their lengths are two modulo four. The initial transcript input has length three modulo four. Later inputs have length zero modulo four. A native field or query response therefore has one typed input. The 40-byte capacity-seed input is fixed separately. The initial context contains the request's second 32-byte digest. The field public values contain its first digest.

2. Rooted reconstruction and source flags

A collision-free database gives every full digest a unique recorded preimage. Starting at a typed response input, follow complete predecessor digests to recover its public instance and earlier responses. Expand the recorded source roots to their fixed heights. Missing source positions are zero-completed. The authenticated preprocessing word is public. This gives the exact rooted prefix used by the source extractor.

A field flag is its finite exceptional response set. A query flag is a passing grind and a subset contained in the prefix's good-position set when source extraction fails. Prefix malformedness and missing final verification pairs are covered by the hash recording term. Each well-formed candidate has the conditional density supplied by the source lemma, uniformly for every fixed earlier database and source word.

The bad property is the presence of at least one such flag for an identifier outside the requested set J. An identifier is the full pair of canonical request digests. It enters J before an authorization is constructed. A terminal fresh accepted proof outside the bad property provides an authority preimage by the source extraction lemma.

3. Global protection of a database prefix

For a parallel query tuple x, delete its addressed rows and fix the remaining database. Collect all recorded digest labels and every typed hash pointer in the remaining keys and in x. Add the fixed public seed label and self-return exclusions. At most beta=8T+32 labels are needed. This set is fixed throughout that tuple's update.

Replace the uniform compression reflection by the reflection using only allowed labels and fixing each forbidden label. If a fraction f is forbidden, the uniform vectors have inner product sqrt(1-f). Their reflection difference is at most 2sqrt(f). The corresponding lookup sandwiches differ by at most 4sqrt(f). Applying the scalar telescoping identity to all physical calls gives global state error at most

```
    E_protect(T)=4T sqrt((8T+32)/N).
```

This comparison is made once for the whole experiment. It is added to the terminal success amplitude after the transition bound.

Every already rooted dependency has an output in the forbidden set. Its compressed row is fixed, including against deletion to the blank symbol. A new row cannot complete an old missing dependency, since its new output would have to equal a forbidden pointer. Thus each earlier rooted prefix is invariant. For one parallel tuple, decompose into the orthogonal fibers of the fixed rooted labels. On each fiber a new flag can arise only at one of its addressed response rows. Its bad set depends only on that row's new output and the fixed prefix.

These are the one-local recognizing sets of CFHL21, Definition 5.14. The uniform distribution on allowed labels increases any density by at most 1/(1-beta/N). The Frobenius-norm proof of CFHL21, Theorem 5.17, applied to the restricted compression reflection therefore gives

```
    transition_s <= sqrt(10 sum_j k_sj kappa_j/(1-beta/N)).
```

The direct-sum fibers do not multiply this bound by their number. The query response is one row, so its entire subset has the joint density.

4. Adaptive constructors and retained input exchanges

The witness-free simulator of simulation.txt uses a classical input permutation whose endpoints are private leaf inputs. It fixes every transcript, internal node and public preprocessing input. A constructor's own identifier is already in J. Its transcript calls do not create a fresh-identifier source flag.

A previous fresh prefix can be changed by a retained exchange only if it uses an old endpoint or a new constructor label. For an old database of at most T rows, a fresh actual endpoint has membership probability at most T/p^{20}. A dummy endpoint conditioned on its digest has membership probability at most $2TN/p^{36}$. A fresh constructor digest hits one of the old pointer labels with probability at most 2(T+v)/N. These bounds hold on arbitrary coherent old states by conditioning on each deleted-row fiber and applying the projector norm bound 2sqrt(probability).

There are at most v endpoints or labels in one attempt. Summing the two comparison directions, raw guard projections and S adaptive attempts is covered, for T>=v, by the following state-distance envelope:

```
E_updates(S,T)=128 S(T+v) sqrt(v/p^20)
    +128 S v sqrt((T+v)/N)
    +128 S v sqrt(2(T+v)N/p^36).
```

The first term dominates the summed actual-endpoint bound $8Sv\ \mathrm{sqrt}(T/p^{20})$, since T>=v. The last two terms cover dummy endpoints and all new-label tests directly. On the complementary fibers the input exchanges preserve all previous fresh prefixes exactly. Pruning and the algebraic-view hybrid are charged separately in eps_sim. The comparison uses raw tapes and aborts the complete experiment on a failed retained guard.

5. Recording and terminal extraction

Count every setup, constructor, projection and final recheck in T. Ordinary compressed-oracle collision search has success amplitude at most 2e(T+1)sqrt(10(T+1)/N)+sqrt(2/N), by CFHL21, Theorem 5.29. Input-label exchanges preserve equality of recorded outputs. Constructor calls are ordinary physical calls. The displayed expression therefore continues to hold between their input-label permutations. Together with at most 65536 terminal pair checks and fixed-row exclusions, use

```
E_record(T)=32 sqrt((T+1)^3/N)+256/sqrt(N).
```

Measure the compressed database once at the end. Apply the final input permutation, enumerate its rooted prefixes and run the source extractor. Validate each returned authority preimage. The extractor has the original quantum execution followed by deterministic classical list decoding. Its complete classical decoding cost is part of the preimage adversary B. It makes no additional quantum queries or rewinds.

6. Resource theorem

For batch s, let k_sj count its responses of type j. A_s is its logical gate count and delta_s its logical depth. The declared batch response price c_j requires sum_j k_sj*c_j <= A_s*delta_s. Let sum A_s<=G and sum delta_s<=D. Cauchy-Schwarz gives

```
sum_s sqrt(sum_j k_sj*kappa_j)
  <= sqrt(GD max_j(kappa_j/c_j)).
```

Put R=GD max_j(kappa_j/c_j), beta=8T+32 and beta<N. The native same-run authorization reduction is

```
    Pr[ReleaseForge] <= Adv_pre(B)+eps_der+eps_rng+eps_sim(S,T)
       +(sqrt(10R/(1-beta/N))+E_protect(T)
          +E_updates(S,T)+E_record(T))^2.
```

The bound is capped at one. Projection from either hash component is simulated by returning the complete hybrid response. Its price premise uses the cheaper component interface. Classical attacks use the same source recognizing sets with the ordinary union bound sum k_sj*kappa_j.

Reference: Chung, Fehr, Huang and Liao, On the Compressed-Oracle Technique, and Post-Quantum Security of Proofs of Sequential Work, https://arxiv.org/abs/2010.11658, Theorems 5.7, 5.17 and 5.29.

Appendix F. Exact host acceptance and hash binding

Every Merkle root compares both complete 32-byte components by equality. The field encoding is injective and every limb is checked below p. A root match therefore requires agreement of SHA256 and Keccak256 on the same typed input. The ideal interface is one uniform 512-bit function with coherent access to either projection. Giving both projections on every query strengthens that adversary. The resource floor is charged at the cheaper component interface.

Canonical requests bind their full SHA256||Keccak256 pair. The first 32 bytes enter the AIR as sixteen u16 limbs. The second component is part of the challenger's initial state. It is recomputed from the same request encoding by each vault. EVM jobs use a two-component key and authenticate both components of the retained cache digest. Solana jobs retain both components of the base and request digests in program-owned state. The nonce is consumed before any value is released.

An unequal base, cache, canonical request or Merkle input accepted under the same retained hybrid digest gives a collision in this 512-bit interface. Source reconstruction also uses the physical input and output pairs. The terminal recording and collision envelope is included in compiler.txt as E_record(T). There is one global hash experiment.

The fixed authority commitment remains 32 bytes. Its target is the already installed vault authority. Recovering an authority preimage is the separate primitive game in authorization.txt. The parameter manifest and preprocessing root are fixed by the deployed verifier.

Solana transports a 64-position vector with its base. Its expected rank is derived from the one native query response. Consecutive verification windows add each position's exact binomial rank term once. At position 64, equality with the expected rank is required before the nonce update and payout. Strict increasing order and the 4096-position range are checked with those terms. The colexicographic rank is a bijection on 64-element subsets. These predicates have exactly the same conjunction as native unranking. Their order creates no additional accepting proof.

Lossless path deltas reconstruct each ordinary query from the previous verified query. At a path divergence, the omitted sibling is the previous leaf hashed through its preceding siblings. Literal siblings replace the other selected components. Every reconstructed path, leaf and fold value passes the ordinary checker. The relayer cannot modify the program-owned cache or advance its contiguous query and rank counters.

Appendix G. Authority primitive conditions

The authorization reduction uses the exact fixed authority-preimage game of authorization.txt, the joint HKDF game and the joint private mask-and-salt generator game. Their advantages are separate terms in the full resource corollary.

Poseidon2 uses KoalaBear p=2130706433, width 24, fourteen secret elements, ten capacity elements and twelve output elements. The capacity is fixed by H(AUTH_DOMAIN). The permutation has eight full rounds and 26 partial rounds. The official Grain generator and round inequalities are reproduced by poseidon_parameters.py. The reference table agrees in all 215 canonical 8/23 constants before generation of the selected 8/26 constants. The selected round count satisfies the generator's classical target-256 inequalities and its margin rules.

The concrete authority condition is preimage resistance for the full installed commitment under the exact fixed constants, capacity and secret distribution. Its resource budget includes the complete extractor described in authorization.txt. The cryptanalysis, matrix checks and round generator support this condition with reproducible evidence. The Category 1 certificate records its required advantage margin explicitly.

One Parano1d secret is 256 bits. HKDF-SHA256 separates networks and account indices. Fourteen 32-bit words are reduced into F_p. The exact residue distribution has either two or three word preimages per field element. This distribution is preserved in the authority game.

Fresh 256-bit operating-system randomness seeds each proof. Distinct SHA256-labelled branches feed the field-mask generator, the Merkle salt generator and the free-row generator. Their joint output is replaced by independent uniform field coins in the private-generator hybrid. The simulator uses twenty independent field salts per matrix row in that hybrid. Public preprocessing has its own deterministic seed.

check_profile.py proves primality by Proth's certificate, irreducibility of X^8-3 by Rabin's criterion and bijectivity of the cube on F_p. It checks both linear-layer ranks, 69 nonzero base-block minors, 96 internal-layer observability powers, the order-64 rotation and all finite mask budgets. check_mask_transport.py checks the exact production-size polynomial translations, all eight quotient cosets and the full aggregate identity. check_rs_certificate.py checks five finite interpolation certificates and thirty native Fourier-fold basis identities.

References https://github.com/HorizenLabs/poseidon2/blob/main/poseidon2_rust_params.sage https://eprint.iacr.org/2023/323 https://eprint.iacr.org/2024/1037 https://eprint.iacr.org/2026/2056

Appendix H. Response-cost condition

The Category 1 corollary uses logical bounded-fanin gates and logical circuit depth. It fixes at most 2^{64} classical authorization requests. For each quantum batch s, with A_s gates, delta_s depth and k_sj newly charged responses of type j, its resource condition is

```
sum_j k_sj c_j <= A_s delta_s,
c_j >= 2^20 gate-depth.
```

The conservative physical call count charges one gate per adversarial call, so adversarial calls are bounded by G. The batch price applies to the cheaper hash component, including parallel execution, workspace reuse and coherent lookup costs. Input and output projections retain this same declared floor.

The resource corollary applies to batches satisfying this minimum-cost premise. The numerical evaluator records the complete success envelope as a function of the price, including a unit-price row and sensitivity.

The concrete assessment in nist-resource-assessment.txt costs published direct-call layouts and native fixed-prefix implementations in a stated logical gate library. check_hash_circuits.py reproduces the native counts, including output projections and a one-bit phase implementation. Its data-only charge favors the attacker by omitting phase, cleanup and fanout costs. The evaluated layouts support the selected reference price. The batch condition above specifies the hypothesis for applying the resource corollary to other circuit or quantum-memory implementations.

The ideal proof-layer calculation and the fixed authority-preimage, derivation and private-generator games are separate inputs. The full Category 1 conclusion applies when their combined advantage is below the remaining half-success margin displayed in security-evaluation.json.

The NIST reference is AES-128 key search. Its quantum reference is $G*D=2^{170}$ at D in $\{2^{40},2^{64},2^{96}\}$. Its classical reference is 2^{143} gates. The resource theorem uses the complete bound including finite simulation, recording and adaptive-update terms at those budgets.

https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization/evaluation-criteria/security-(evaluation-criteria)